Holographic RIS-Aided Reference Energy Detection for Low-Complexity Noncoherent Communications

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***Abstract*—This paper presents a comprehensive framework for signal recovery in holographic reconfigurable intelligent surface (RIS) receivers, where only amplitude information is available at the receiver. The proposed system employs a known reference wave to enable phase retrieval from intensity-only measurements. We formulate the complete signal model from transmission to reception, and propose a Gerchberg-Saxton algorithm to estimate the wave vector and complex channel gain. Building upon this, we develop a Gauss-Newton algorithm for efficient signal recovery to conduct real-time estimation of the symbols utilizing the property of rapid convergence. Simulation results validate the effectiveness of the proposed approach, showing accurate signal recovery under various signal-to-noise ratio conditions. The complete modeling framework and algorithm implementation details are provided to facilitate reproducibility and further research in RIS-based communication systems.**

***Index Terms*—Reconfigurable Intelligent Surface, holographic receiver, phase retrieval, signal recovery, amplitude-only measurements.**

I. Introduction

Reconfigurable intelligent surfaces (RISs) have emerged as a promising technology for next-generation wireless communication systems, owing to their ability to reshape the wireless propagation environment in a programmable and energy-efficient manner [1], [2]. By adaptively controlling the electromagnetic response of a large number of low-cost reflecting elements, RISs can provide additional degrees of freedom for coverage enhancement, interference management, and spatial signal manipulation. More recently, the functionality of RISs has been extended beyond passive beamforming toward active reception and signal processing, giving rise to the concept of holographic RIS receivers [3]. By exploiting densely deployed metasurface elements, holographic RIS receivers are expected to capture rich spatial information of impinging electromagnetic waves while maintaining much lower hardware complexity than conventional antenna arrays equipped with fully digital radio-frequency chains.

Despite these advantages, a fundamental bottleneck of holographic RIS receivers lies in the acquisition of complex-valued signal information. Conventional coherent receivers rely on down-conversion, local oscillator synchronization, and in-phase/quadrature sampling to obtain both amplitude and phase information. In contrast, practical RIS-based or metasurface-based receivers may only have access to intensity or power measurements due to hardware, power, and integration constraints [4]. As a result, the received signal phase is not directly observable, and the recovery of complex-valued symbols from intensity-only measurements naturally leads to a phase retrieval problem, which has been extensively studied in optics and signal processing [5], [6].

A promising way to alleviate this ambiguity is to introduce a known reference wave before intensity detection. Inspired by optical holography, where a reference beam interferes with an object wave to encode both amplitude and phase information into an intensity pattern [7], a reference-assisted RIS receiver superimposes a known radio-frequency reference signal with the unknown received signal at the metasurface or receiver front end. The resulting square-law measurement contains not only the power of the unknown signal but also a reference-induced interference term. This interference term serves as a phase anchor and makes it possible to infer the hidden complex-valued signal from real-valued intensity observations.

However, existing studies on RIS-assisted reception and phase retrieval still leave several key questions insufficiently addressed. Most RIS receiver designs focus on enhancing the received signal power, improving beamforming gain, or reducing hardware complexity, while the intrinsic identifiability of complex symbols under intensity-only measurements has not been fully characterized. In particular, when the receiver observes only the squared magnitude of the superimposed field, different transmitted symbols may produce identical or nearly identical intensity patterns. In such cases, no detector, including maximum-likelihood detection or iterative phase retrieval algorithms, can reliably distinguish the transmitted symbols. Therefore, the central issue is not only how to design an efficient recovery algorithm, but also whether the transmitted symbols are uniquely encoded in the intensity domain in the first place.

This issue becomes more critical in holographic RIS receivers because the measured intensity pattern is jointly determined by multiple coupled components, including the reference wave, the pulse-shaping or sampling operation, and the RIS array manifold. If the effective matrix maps two different symbol vectors to the same field response, the corresponding

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intensity measurements remain indistinguishable regardless of how the detector is implemented. Likewise, even when the effective field responses are different, an improperly designed reference wave may fail to project their phase difference into the observable intensity domain.

Consequently, the lack of a systematic identifiability analysis limits the theoretical understanding and practical design of reference-assisted holographic RIS receivers. Existing phase retrieval methods often assume generic random measurements or focus on continuous signal recovery, whereas communication symbols belong to finite constellations with specific algebraic structures. This discrete nature creates both opportunities and challenges: on one hand, exact recovery may be possible under weaker conditions than those required for general continuous phase retrieval; on the other hand, constellation-dependent ambiguities may still arise if the reference wave and the RIS measurement structure are not properly designed. Therefore, it is essential to investigate how the reference wave, the shaping matrix, and the RIS array manifold jointly determine the uniqueness and stability of symbol recovery.

In this work, we address this problem by developing a reference-assisted phase retrieval framework for holographic RIS receivers. Instead of treating the reference signal merely as an auxiliary hardware component, we characterize its role as a phase-anchoring mechanism that converts hidden phase information into measurable intensity variations. We analyze the discrete identifiability condition of transmitted symbols under the reference-assisted intensity observation model and show that successful recovery requires two complementary properties: the effective RIS measurement matrix must preserve the differences between constellation points, and the reference wave must separate these differences in the intensity domain. Based on this insight, we further introduce an energy-domain minimum distance metric to quantify the distinguishability of different transmitted symbols after square-law detection. This metric provides a direct link between reference-wave design, RIS measurement diversity, and symbol detection performance.

The main contributions of this paper are summarized as follows:

- We establish a reference-assisted holographic RIS receiver model, in which complex-valued communication symbols are recovered from intensity-only measurements through the interference between the unknown signal and a known reference wave.
- We derive discrete identifiability conditions for reference-assisted phase retrieval over finite constellations and reveal the joint impact of the reference wave, the pulse-shaping matrix, and the RIS array manifold on symbol uniqueness.
- We introduce an energy-domain minimum distance metric to characterize the stability of symbol recovery and to guide reference-wave and RIS measurement design.
- We develop and compare maximum-likelihood and low-complexity phase retrieval based detectors, demonstrating that proper reference-wave diversity significantly improves symbol distinguishability and detection performance.

II. System Model

We consider a holographic RIS receiver system in which L single-antenna users transmit complex baseband symbols to an RIS composed of K reflecting elements. The RIS senses only the intensity of the interference pattern formed by the received signal and a known reference wave at each of its elements. The known reference signal is generated by a local oscillator [8]. The goal is to estimate the complex channel parameters and degree of arrival (DOA) angles, and subsequently recover the transmitted symbols utilizing these channel information.

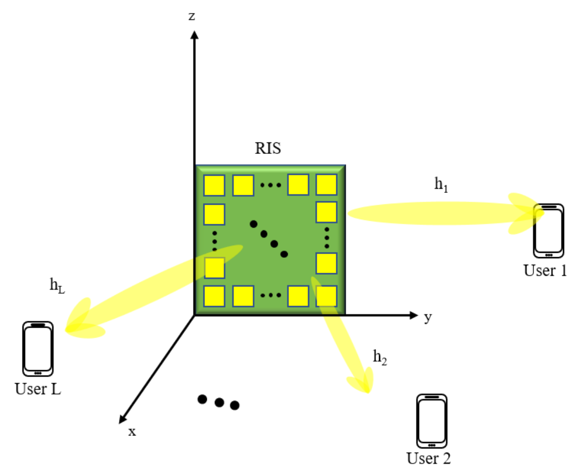

Fig. 1: System model of holographic RIS receiver with amplitude-only measurements.

A. Transmitter Model

Consider an uplink transmission system with L single-antenna users. The information-bearing symbols transmitted by the l-th user within one signaling block are collected in

$$\mathbf{s}_l = [s_{l,0}, s_{l,1}, \ldots, s_{l,K-1}]^T \in \mathcal{S}^K, \tag{1}$$

where K denotes the number of symbols in the considered block, $s_{l,k}$ is the k-th transmitted symbol of user l, and $\mathcal{S}$ is the modulation constellation. The constellation symbols are assumed to be normalized as

$$\mathbb{E}\{|s_{l,k}|^2\} = 1. \tag{2}$$

After pulse shaping, the complex baseband transmit waveform of the l-th user is given by

$$x_l(t) = \sqrt{P_l}\left(\sum_{k=0}^{K-1} s_{l,k}\, p(t - kT_s) + \xi_l q(t)\right), \tag{3}$$

where P_l is the average transmit power of user l, $p(t)$ is the pulse-shaping filter, T_s is the symbol duration, $\xi_l \in \mathbb{C}$ is a deterministic pilot or bias coefficient, and $q(t)$ is a known pilot waveform. The deterministic term $\xi_l q(t)$ is optional and is introduced to allow a nonzero-mean transmitted waveform when required by the subsequent reference-assisted recovery process. If no transmitted pilot or bias is used, we set $\xi_l = 0$.

The corresponding real-valued passband signal is expressed as

$$x_l^{\mathrm{RF}}(t) = \sqrt{2}\Re\left\{x_l(t)e^{j(2\pi f_c t+\varphi_l)}\right\}, \quad l = 1, 2, \ldots, L, \quad (4)$$

where f_c is the carrier frequency and φ_l denotes the initial carrier phase offset of the l-th user. Let

$$x_l(t) = x_{l,I}(t) + jx_{l,Q}(t), \quad (5)$$

where $x_{l,I}(t)$ and $x_{l,Q}(t)$ are the in-phase and quadrature components of the complex baseband waveform, respectively. Then (4) can be equivalently written as

$$x_l^{\mathrm{RF}}(t) = \sqrt{2}\left[x_{l,I}(t)\cos(2\pi f_c t + \varphi_l) - x_{l,Q}(t)\sin(2\pi f_c t + \varphi_l)\right]. \quad (6)$$

For compact notation, define the multi-user symbol vector at symbol time k as

$$\mathbf{s}_k = [s_{1,k}, s_{2,k}, \ldots, s_{L,k}]^T \in \mathcal{S}^L, \quad (7)$$

and the multi-user complex baseband waveform vector as

$$\mathbf{x}(t) = [x_1(t), x_2(t), \ldots, x_L(t)]^T \in \mathbb{C}^L. \quad (8)$$

Using (3), the multi-user baseband waveform can be written as

$$\mathbf{x}(t) = \mathbf{D}_P\left(\sum_{k=0}^{K-1}\mathbf{s}_k p(t - kT_s) + \boldsymbol{\xi} q(t)\right), \quad (9)$$

where

$$\mathbf{D}_P = \mathrm{diag}(\sqrt{P_1}, \sqrt{P_2}, \ldots, \sqrt{P_L}) \quad (10)$$

is the transmit-power scaling matrix, and

$$\boldsymbol{\xi} = [\xi_1, \xi_2, \ldots, \xi_L]^T \in \mathbb{C}^L \quad (11)$$

collects the deterministic pilot or bias coefficients of all users.

To facilitate the subsequent symbol-level analysis, we further introduce an equivalent matrix representation for the pulse-shaping operation. Consider N_t baseband observation instants $\{t_n\}_{n=0}^{N_t-1}$ within the signaling block. For the l-th user, define the sampled baseband transmit vector as

$$\tilde{\mathbf{x}}_l = [x_l(t_0), x_l(t_1), \ldots, x_l(t_{N_t-1})]^T \in \mathbb{C}^{N_t}. \quad (12)$$

From (3), we obtain

$$\tilde{\mathbf{x}}_l = \sqrt{P_l}\left(\mathbf{P}\mathbf{s}_l + \xi_l\mathbf{q}\right), \quad (13)$$

where $\mathbf{P} \in \mathbb{C}^{N_t\times K}$ is the pulse-shaping matrix with entries

$$[\mathbf{P}]_{n,k} = p(t_n - kT_s), \quad n = 0, 1, \ldots, N_t - 1, \quad k = 0, 1, \ldots, K-1, \quad (14)$$

and

$$\mathbf{q} = [q(t_0), q(t_1), \ldots, q(t_{N_t-1})]^T \in \mathbb{C}^{N_t} \quad (15)$$

is the sampled pilot waveform.

Stacking the sampled baseband transmit vectors of all users gives

$$\tilde{\mathbf{x}} = [\tilde{\mathbf{x}}_1^T, \tilde{\mathbf{x}}_2^T, \ldots, \tilde{\mathbf{x}}_L^T]^T \in \mathbb{C}^{LN_t}. \quad (16)$$

Similarly, define the stacked transmitted symbol vector as

$$\mathbf{s} = [\mathbf{s}_1^T, \mathbf{s}_2^T, \ldots, \mathbf{s}_L^T]^T \in \mathcal{S}^{LK}. \quad (17)$$

Then the sampled multi-user transmit waveform can be represented as

$$\tilde{\mathbf{x}} = (\mathbf{D}_P \otimes \mathbf{P})\mathbf{s} + (\mathbf{D}_P\boldsymbol{\xi}) \otimes \mathbf{q}, \quad (18)$$

where $\otimes$ denotes the Kronecker product. For notational simplicity, define

$$\mathbf{P}_{\mathrm{tx}} = \mathbf{D}_P \otimes \mathbf{P} \in \mathbb{C}^{LN_t\times LK} \quad (19)$$

and

$$\mathbf{x}_0 = (\mathbf{D}_P\boldsymbol{\xi}) \otimes \mathbf{q} \in \mathbb{C}^{LN_t}. \quad (20)$$

Thus, the transmitter-side equivalent discrete model is compactly written as

$$\tilde{\mathbf{x}} = \mathbf{P}_{\mathrm{tx}}\mathbf{s} + \mathbf{x}_0. \quad (21)$$

The matrix $\mathbf{P}_{\mathrm{tx}}$ characterizes the linear mapping from the transmitted constellation symbols to the sampled pulse-shaped waveforms. This representation allows the subsequent receiver model to incorporate pulse shaping, spatial propagation, and holographic RIS observations through an equivalent measurement matrix while keeping the analysis in the discrete symbol domain.

B. Channel Model

As shown in Figure 1, we consider a holographic RIS receiver consisting of an $N \times M$ array of elements placed on the yz -plane, for a total of $K = NM$ receiving elements.

Noting that the RIS is placed on the y-z plane, the position vector of (n, m) th RIS element is $\mathbf{r}_{n,m} = [0, y_n, z_m]^T$, where $y_n = (n-1)d_y$ and $z_m = (m-1)d_z$. Here, d_y and d_z are the spacing between adjacent elements along the y-axis and z-axis, respectively. The index (n, m) represents the n-th row and m-th column of the unit.

We define wave number vector of each user $\mathbf{k_l} = [k_{l,x}, k_{l,y}, k_{l,z}]^\top$, where $k_{l,x} = k\cos\theta_l$, $k_{l,y} = k\sin\theta_l\cos\phi_l$, and $k_{l,z} = k\sin\theta_l\sin\phi_l$. Here, θ_l and ϕ_l are the elevation and azimuth angles of arrival for user l, respectively, and $k = \frac{2\pi}{\lambda}$ is the wave number with λ being the wavelength. Thus the (n, m)-th RIS element experiences a phase shift of $\psi_{n,m}^{(l)} = \mathbf{k_l}^T\mathbf{r}_{n,m} = k_{l,y}y_n + k_{l,z}z_m$ due to the incident wave from user l.

Following [9], for each user, the incident signal across the RIS elements is assumed to have uniform amplitude, with variations only in phase; however, both amplitude and phase may differ across users. Thus we can express the vector $\mathbf{h_l}$ for l-th user with the two directional array steering vector as

$$\begin{aligned}\mathbf{h_l} &= \alpha_l\mathbf{a_{l,y}}(\theta, \phi) \otimes \mathbf{a_{l,z}}(\theta, \phi) \\ &\triangleq \alpha_l\mathbf{v_l}, \in \mathbb{C}^{K\times 1}\end{aligned} \quad (22)$$

where α_l is the complex channel gain for user l, and $\otimes$ denotes the Kronecker product. $\mathbf{a_{l,y}}(\theta, \phi)$ and $\mathbf{a_{l,z}}(\theta, \phi)$ are the array steering vectors along the y-axis and z-axis for user l, respectively, defined as

$$\mathbf{a_{l,y}}(\theta, \phi) = [e^{jk_{l,y}y_1}, e^{jk_{l,y}y_2}, \ldots, e^{jk_{l,y}y_N}]^\top \quad (23)$$

$$\mathbf{a_{l,z}}(\theta,\phi)=[e^{jk_{l,z}z_1},e^{jk_{l,z}z_2},\ldots,e^{jk_{l,z}z_M}]^\top \tag{24}$$

Then for the all users, the channel matrix can be expressed as

$$\begin{aligned}\mathbf{H}&=[\mathbf{h_1},\mathbf{h_2},\ldots,\mathbf{h_L}]\\&=[\alpha_1\mathbf{v_1},\alpha_2\mathbf{v_2},\ldots,\alpha_L\mathbf{v_L}]\\&\triangleq\mathbf{V}\cdot\mathbf{A}\in\mathbb{C}^{K\times L}\end{aligned} \tag{25}$$

where $\mathbf{V}=[\mathbf{v_1},\mathbf{v_2},\ldots,\mathbf{v_L}]\in\mathbb{C}^{K\times L}$ is the array steering matrix for all users, and $\mathbf{A}=\mathrm{diag}(\alpha_1,\alpha_2,\ldots,\alpha_L)\in\mathbb{C}^{L\times L}$ is the diagonal matrix of complex channel gains.

C. Received Signal Model

At the receiver side, a local oscillator (LO) is employed to generate multiple reference wave signals with known amplitudes and phases. The i-th reference signal in complex baseband form is expressed as

$$c_i=A_ie^{j\phi_i},\quad i=1,2,\ldots,N, \tag{26}$$

where A_i and ϕ_i denote the amplitude and phase of the i-th reference signal, respectively.

The corresponding passband RF reference signal is given by

$$a_i(t)=\Re\left\{c_ie^{j2\pi f_ct}\right\}=A_i\cos(2\pi f_ct+\phi_i). \tag{27}$$

The received RF signal at the i-th antenna is then superimposed with the reference signal, yielding

$$y_i(t)=r_i(t)+a_i(t), \tag{28}$$

where $r_i(t)$ denotes the received RF signal defined previously.

Then, the received RF signal at the i-th RIS element from all L users is given by

$$g_i(t)=\sum_{l=1}^{L}\Re\left\{h_{i,l}s_l(t)e^{j2\pi f_ct}\right\}+a_i(t)+n_i(t), \tag{29}$$

where $s_l(t)$ denotes the complex baseband signal of the l-th user after pulse shaping, and $h_{i,l}$ represents the channel coefficient between the l-th user and the i-th RIS element.

Under the far-field assumption, the channel can be modeled as

$$h_{i,l}=\alpha_lv_{i,l}e^{j\phi_l}, \tag{30}$$

where α_l denotes the large-scale fading coefficient, ϕ_l is the initial phase offset, and $v_{i,l}$ is the (i,l)-th element of the array steering matrix $\mathbf{V}$.

Therefore, the received signal can be rewritten as

$$\begin{aligned}g_i(t)=&\sum_{l=1}^{L}\Re\left\{\alpha_lv_{i,l}e^{j\phi_l}s_l(t)e^{j2\pi f_ct}\right\}\\&+A_i\cos(2\pi f_ct+\phi_{b,i})+n_i(t).\end{aligned} \tag{31}$$

We stack the received RF signals at all N RIS elements into a vector form as

$$\mathbf{g}(t)=\left[g_1(t),g_2(t),\ldots,g_N(t)\right]^T. \tag{32}$$

Based on the signal model in the previous section, the received signal can be expressed as

$$\mathbf{g}(t)=\Re\left\{\mathbf{H}\mathbf{s}(t)e^{j2\pi f_ct}\right\}+\mathbf{a}(t)+\mathbf{n}(t), \tag{33}$$

where $\mathbf{s}(t)\in\mathbb{C}^{L\times1}$ is the transmitted baseband signal vector, and $\mathbf{H}\in\mathbb{C}^{N\times L}$ denotes the channel matrix.

Under the far-field assumption, the channel matrix can be decomposed as

$$\mathbf{H}=\mathbf{V}\mathbf{A}, \tag{34}$$

where $\mathbf{V}\in\mathbb{C}^{N\times L}$ is the array steering matrix, and $\mathbf{A}=\mathrm{diag}(\alpha_1e^{j\phi_1},\ldots,\alpha_Le^{j\phi_L})$ contains the complex channel gains.

After square-law detection and time averaging, the received signal intensity at all RIS elements is given by

$$\mathbf{z}=\left[z_1,z_2,\ldots,z_N\right]^T, \tag{35}$$

where

$$z_i=\frac{1}{T_0}\int_{T_0}|g_i(t)|^2dt. \tag{36}$$

where T_0 is the integration time interval for square-law detection.

Under the narrowband and slowly varying envelope assumptions, the measurement can be approximated as

$$\mathbf{z}\approx\left|\mathbf{H}\mathbf{s}+\mathbf{c}\right|^2+\mathbf{w}, \tag{37}$$

where $\mathbf{c}\in\mathbb{C}^{N\times1}$ denotes the equivalent complex reference vector determined by $\mathbf{a}(t)$, $\mathbf{w}$ represents the effective noise after square-law detection, and $|\cdot|^2$ is applied element-wise.

III. Problem Formulation

The channel estimation and signal recovery problem can be formulated as estimating the complex signal s, the complex channel gain $\tilde{\alpha}$ the azimuth angle ϕ and elevation angle θ from the intensity measurements $\mathbf{y}$, given the known reference wave b.

In order to deduce the complex signal s from the intensity measurements $\{I_i\}_{i=1}^K$, we need to firstly estimate the complex channel h_i and then estimate the complex signal s. This can be achieved by a two-stage estimation framework. In the first stage, we use pilot symbols to estimate the channel parameters, including the complex channel gains α_l and the DOA angles θ_l and ϕ_l for each user l. In the second stage, we utilize the estimated channel parameters to recover the transmitted symbols from the intensity measurements during data transmission.

A. Stage I: Channel Parameter Estimation

Assuming that the channel remains constant within a coherence interval, we employ known pilot symbols to estimate the channel parameters. Let $\{s_{l,t}\}_{t=1}^{T_p}$ denote the pilot symbols transmitted by user l during the pilot phase, where T_p is the number of pilot symbols.

Define the pilot symbol vector at time index t as

$$\mathbf{s}_t=\left[s_{1,t},s_{2,t},\ldots,s_{L,t}\right]^T\in\mathbb{C}^{L\times1}. \tag{38}$$

Based on the intensity measurement model in (37), the received signal intensity at the RIS array during the pilot phase can be expressed as

$$\mathbf{z}_t \approx |\mathbf{H}\mathbf{s}_t + \mathbf{c}|^2 + \mathbf{w}_t, \tag{39}$$

where $\mathbf{H} = \mathbf{V}\mathbf{A} \in \mathbb{C}^{N \times L}$ is the channel matrix, $\mathbf{V} \in \mathbb{C}^{N \times L}$ is the array steering matrix determined by the angles of arrival (θ_l, ϕ_l), and $\mathbf{A} = \mathrm{diag}(\alpha_1 e^{j\phi_1}, \ldots, \alpha_L e^{j\phi_L})$ contains the complex channel gains.

The vector $\mathbf{c} \in \mathbb{C}^{N \times 1}$ denotes the equivalent reference signal, and $\mathbf{w}_t$ represents the effective noise after square-law detection. The operation $|\cdot|^2$ is applied element-wise.

Collecting all pilot observations, we obtain

$$\mathbf{Z} = \left[\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_{T_p}\right]. \tag{40}$$

During the pilot phase, the transmitted symbols $\{s_{l,t}\}$ are known at the receiver. Define the pilot symbol vector at time t as

$$\mathbf{s}_t = [s_{1,t}, s_{2,t}, \ldots, s_{L,t}]^T. \tag{41}$$

We define the unknown parameter vector as

$$\boldsymbol{\eta} = \left[\boldsymbol{\theta}^{\mathsf{T}}, \boldsymbol{\phi}^{\mathsf{T}}, \Re\{\boldsymbol{\alpha}\}^{\mathsf{T}}, \Im\{\boldsymbol{\alpha}\}^{\mathsf{T}}\right]^{\mathsf{T}} \in \mathbb{R}^{4L}, \tag{42}$$

where $\boldsymbol{\theta} = [\theta_1, \ldots, \theta_L]^T$, $\boldsymbol{\phi} = [\phi_1, \ldots, \phi_L]^T$, and $\boldsymbol{\alpha} = [\alpha_1, \ldots, \alpha_L]^T$ denotes the complex channel gains.

The channel matrix is parameterized as

$$\mathbf{H}(\boldsymbol{\eta}) = \mathbf{V}(\boldsymbol{\theta}, \boldsymbol{\phi})\, \mathrm{diag}(\boldsymbol{\alpha}), \tag{43}$$

where $\mathbf{V}(\boldsymbol{\theta}, \boldsymbol{\phi}) \in \mathbb{C}^{N \times L}$ is the array steering matrix determined by the angles of arrival.

The pilot-based channel estimation problem is formulated as a nonlinear least-squares (NLS) problem:

$$\min_{\boldsymbol{\eta}} \sum_{t=1}^{T_p} \left\| \mathbf{z}_t - \boldsymbol{\mu}_t(\boldsymbol{\eta}) \right\|_2^2, \tag{44}$$

where the model output is given by

$$\boldsymbol{\mu}_t(\boldsymbol{\eta}) = \left|\mathbf{H}(\boldsymbol{\eta})\mathbf{s}_t + \mathbf{c}\right|^2, \tag{45}$$

and $|\cdot|^2$ is applied element-wise.

The nonlinear measurement model in (39) can be reformulated as a phase retrieval problem. Define

$$\mathbf{y}_t = \mathbf{H}(\boldsymbol{\eta})\mathbf{s}_t + \mathbf{c}, \tag{46}$$

then the intensity measurements can be written as

$$\mathbf{z}_t = |\mathbf{y}_t|^2. \tag{47}$$

Since only the magnitude of $\mathbf{y}_t$ is observed while its phase is lost, recovering $\mathbf{y}_t$ from $\mathbf{z}_t$ constitutes a classical phase retrieval problem.

To solve this problem, we adopt the Gerchberg–Saxton (GS) algorithm, which iteratively enforces consistency between the measured magnitude and the structural constraint imposed by the channel model.

Specifically, let $\mathbf{y}_t^{(k)}$ denote the estimate of $\mathbf{y}_t$ at the k-th iteration. The GS algorithm consists of the following two steps:

1) Magnitude Projection:

$$\tilde{\mathbf{y}}_t^{(k)} = \sqrt{\mathbf{z}_t} \odot \frac{\mathbf{y}_t^{(k)}}{|\mathbf{y}_t^{(k)}|}, \tag{48}$$

where $\odot$ denotes the element-wise product.

2) Signal Projection:

$$\mathbf{y}_t^{(k+1)} = \mathbf{H}(\boldsymbol{\eta}^{(k)})\mathbf{s}_t + \mathbf{c}, \tag{49}$$

where $\boldsymbol{\eta}^{(k)}$ is updated by solving

$$\boldsymbol{\eta}^{(k)} = \arg\min_{\boldsymbol{\eta}} \sum_{t=1}^{T_p} \left\| \tilde{\mathbf{y}}_t^{(k)} - \mathbf{H}(\boldsymbol{\eta})\mathbf{s}_t - \mathbf{c} \right\|_2^2. \tag{50}$$

The above two steps are repeated until convergence.

B. Stage II: Signal Recovery from Intensity Measurements

During the data transmission phase, $\hat{\mathbf{V}}$ and $\hat{\mathbf{A}}$ are assumed to be known. For each time slot t, the symbol vector $\mathbf{s}_t \in \mathbb{C}^L$ is estimated by solving

$$\min_{\mathbf{s}_t} \left\| \mathbf{I}_t - \left| \hat{\mathbf{V}}\hat{\mathbf{A}}\mathbf{s}_t + \mathbf{b} \right|^2 \right\|_2^2. \tag{51}$$

It's clear that the signal recovery problem in (51) has the same mathematical structure as the channel estimation problem in (44), and can also be solved by the Gauss-Newton algorithm.

We use QPSK modulation for symbol transmission as an example, which is also commonly used in other practical communication systems. Define the real-valued parameterization

$$\mathbf{x}_t = \left[\Re\{\mathbf{s}_t\}^{\mathsf{T}}, \Im\{\mathbf{s}_t\}^{\mathsf{T}}\right]^{\mathsf{T}} \in \mathbb{R}^{2L}. \tag{52}$$

The Gauss–Newton update for symbol estimation is given by

$$\mathbf{x}_t^{(i+1)} = \mathbf{x}_t^{(i)} - \left(\mathbf{J}_s^{\mathsf{T}}\mathbf{J}_s\right)^{-1} \mathbf{J}_s^{\mathsf{T}} \left(\boldsymbol{\mu}_t - \mathbf{I}_t\right), \tag{53}$$

where $\mathbf{J}_s = \partial\boldsymbol{\mu}_t / \partial\mathbf{x}_t$. The final symbol estimate is obtained via hard QPSK decision.

We can summarize the complete two-stage estimation algorithm in Algorithm 1.

IV. CRB Analysis and Theoretical Bounds Evaluation

In this section, we analyze the theoretical performance limits of the proposed two-stage estimation framework by deriving the Cramér–Rao bound (CRB) for both channel parameter estimation and symbol recovery. Further more, we evaluate the error bounds through theoretical analysis.

A. Cramér–Rao Bound Analysis

In this subsection, we derive the Cramér–Rao bound (CRB) for the proposed power-domain estimation framework. The CRB characterizes the minimum achievable mean-squared error (MSE) of any unbiased estimator and serves as a fundamental performance benchmark for the proposed two-stage Gauss–Newton algorithm.

Algorithm 1 Two-Stage Gauss–Newton Estimation for RIS-Assisted System

Input: Power measurements $\{\mathbf{I}_t\}$, pilot symbols $\{\mathbf{s}_t\}_{t=1}^{T_p}$, reference wave $\mathbf{b}$
Initialization: Initial DOA estimates $\boldsymbol{\theta}^{(0)}, \boldsymbol{\phi}^{(0)}$ and channel gains $\boldsymbol{\alpha}^{(0)}$
Stage I: Pilot-Based Estimation
for $i = 0$ to $I_{\max}$ **do**
  Compute residual vector $\mathbf{r}$
  Compute Jacobian matrix $\mathbf{J}$
  Update parameters using (**??**)
end for
Obtain $\hat{\mathbf{V}}$ and $\hat{\mathbf{A}}$
Stage II: Data Symbol Estimation
for each data time slot t **do**
  Initialize $\mathbf{x}_t^{(0)} = \mathbf{0}$
  for $i = 0$ to $I_{\max}$ **do**
    Compute residual $\boldsymbol{\mu}_t - \mathbf{I}_t$
    Compute Jacobian $\mathbf{J}_s$
    Update $\mathbf{x}_t$ using (53)
  end for
  Perform QPSK hard decision to obtain $\hat{\mathbf{s}}_t$
end for
Output: Estimated DOAs $\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}}$, channel gains $\hat{\boldsymbol{\alpha}}$, and data symbols $\{\hat{\mathbf{s}}_t\}$

1) Statistical Model of Power Measurements: Consider the power measurement model at the RIS during the t-th time slot

$$\mathbf{I}_t = \boldsymbol{\mu}_t(\boldsymbol{\eta}) + \mathbf{n}_t, \tag{54}$$

where

$$\boldsymbol{\mu}_t(\boldsymbol{\eta}) = \left| \sum_{l=1}^{L} \alpha_l \mathbf{v}(\theta_l, \phi_l) s_{l,t} + \mathbf{b} \right|^2 \in \mathbb{R}^K, \tag{55}$$

and $\mathbf{n}_t \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_K)$ is additive white Gaussian noise. The unknown parameter vector is defined as

$$\boldsymbol{\eta} = \left[\boldsymbol{\theta}^{\mathsf{T}}, \boldsymbol{\phi}^{\mathsf{T}}, \Re\{\boldsymbol{\alpha}\}^{\mathsf{T}}, \Im\{\boldsymbol{\alpha}\}^{\mathsf{T}}\right]^{\mathsf{T}} \in \mathbb{R}^{4L}. \tag{56}$$

Stacking all pilot observations yields

$$\mathbf{I} = \boldsymbol{\mu}(\boldsymbol{\eta}) + \mathbf{n}, \tag{57}$$

where $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_{KT_p})$.

2) Fisher Information Matrix: Under the Gaussian noise assumption, the Fisher information matrix (FIM) for $\boldsymbol{\eta}$ is given by

$$\mathbf{F}(\boldsymbol{\eta}) = \frac{1}{\sigma^2} \mathbf{J}^{\mathsf{T}}(\boldsymbol{\eta}) \mathbf{J}(\boldsymbol{\eta}), \tag{58}$$

where

$$\mathbf{J}(\boldsymbol{\eta}) = \frac{\partial \boldsymbol{\mu}(\boldsymbol{\eta})}{\partial \boldsymbol{\eta}} \in \mathbb{R}^{KT_p \times 4L} \tag{59}$$

is the Jacobian matrix of the noiseless power measurements with respect to the unknown parameters.

For the t-th pilot snapshot, the Jacobian can be expressed as

$$\mathbf{J}_t = \left[\frac{\partial \boldsymbol{\mu}_t}{\partial \boldsymbol{\theta}}, \frac{\partial \boldsymbol{\mu}_t}{\partial \boldsymbol{\phi}}, \frac{\partial \boldsymbol{\mu}_t}{\partial \Re\{\boldsymbol{\alpha}\}}, \frac{\partial \boldsymbol{\mu}_t}{\partial \Im\{\boldsymbol{\alpha}\}} \right]. \tag{60}$$

By defining

$$\mathbf{y}_t = \sum_{l=1}^{L} \alpha_l \mathbf{v}(\theta_l, \phi_l) s_{l,t} + \mathbf{b}, \tag{61}$$

the k-th element of $\boldsymbol{\mu}_t$ is given by

$$[\boldsymbol{\mu}_t]_k = |[\mathbf{y}_t]_k|^2 = \mathbf{y}_t^{\mathsf{H}} \mathbf{e}_k \mathbf{e}_k^{\mathsf{T}} \mathbf{y}_t, \tag{62}$$

where $\mathbf{e}_k$ is the k-th canonical basis vector.

Using the chain rule, the partial derivative with respect to a generic real parameter η_i is

$$\frac{\partial \boldsymbol{\mu}_t}{\partial \eta_i} = 2\Re \left\{ \operatorname{diag}(\mathbf{y}_t)^{\mathsf{H}} \frac{\partial \mathbf{y}_t}{\partial \eta_i} \right\}. \tag{63}$$

3) CRB for DOA and Channel Gain Estimation: The CRB for the joint estimation of DOAs and channel gains is given by

$$\mathrm{CRB}(\boldsymbol{\eta}) = \mathbf{F}^{-1}(\boldsymbol{\eta}). \tag{64}$$

In particular, the lower bounds for the elevation and azimuth angles of user l are obtained as

$$\mathrm{CRB}(\theta_l) = \left[\mathbf{F}^{-1}(\boldsymbol{\eta})\right]_{l,l}, \quad \mathrm{CRB}(\phi_l) = \left[\mathbf{F}^{-1}(\boldsymbol{\eta})\right]_{L+l,L+l}. \tag{65}$$

These bounds quantify the best achievable accuracy of DOA estimation from power-only measurements in the presence of a known reference wave.

4) Conditional CRB for Symbol Estimation: During the data transmission phase, the steering matrix $\hat{\mathbf{V}}$ and channel gain matrix $\hat{\mathbf{A}}$ are assumed to be known. The symbol estimation problem is governed by

$$\mathbf{I}_t = \boldsymbol{\mu}_t(\mathbf{s}_t) + \mathbf{n}_t, \tag{66}$$

where

$$\boldsymbol{\mu}_t(\mathbf{s}_t) = \left| \hat{\mathbf{V}} \hat{\mathbf{A}} \mathbf{s}_t + \mathbf{b} \right|^2. \tag{67}$$

Define the real-valued symbol parameter vector

$$\mathbf{x}_t = \left[\Re\{\mathbf{s}_t\}^{\mathsf{T}}, \Im\{\mathbf{s}_t\}^{\mathsf{T}}\right]^{\mathsf{T}} \in \mathbb{R}^{2L}. \tag{68}$$

The corresponding Fisher information matrix is

$$\mathbf{F}_s = \frac{1}{\sigma^2} \mathbf{J}_s^{\mathsf{T}} \mathbf{J}_s, \tag{69}$$

where $\mathbf{J}_s = \partial \boldsymbol{\mu}_t / \partial \mathbf{x}_t$. The conditional CRB for symbol estimation is therefore

$$\mathrm{CRB}(\mathbf{s}_t) = \mathbf{F}_s^{-1}. \tag{70}$$

The diagonal entries of $\mathbf{F}_s^{-1}$ provide lower bounds on the MSE of the real and imaginary parts of each user's transmitted symbol. These bounds can be directly related to the achievable symbol error rate (SER) under QPSK modulation.

B. Error Bound Analysis via Perturbation and Inequality Relaxation

While the Cramér–Rao bound characterizes the fundamental statistical limits of unbiased estimation, it does not explicitly quantify how the estimation error behaves for a given algorithm under finite noise realizations. In this subsection, we derive deterministic upper bounds on the estimation error norms by exploiting inequality relaxation and perturbation analysis.

1) Problem Setup and Error Definition: Consider the noiseless power measurement model

$$\mathbf{I}_t^\star = \left| \sum_{l=1}^{L} \alpha_l^\star \mathbf{v}(\theta_l^\star, \phi_l^\star) s_{l,t} + \mathbf{b} \right|^2, \tag{71}$$

where $\{\theta_l^\star, \phi_l^\star, \alpha_l^\star\}$ denote the true parameters. The noisy measurement is given by

$$\mathbf{I}_t = \mathbf{I}_t^\star + \mathbf{n}_t, \qquad \|\mathbf{n}_t\|_2 \le \varepsilon, \tag{72}$$

where ε denotes a noise-dependent bound holding with high probability.

Let $\hat{\boldsymbol{\eta}}$ denote the estimated parameter vector obtained from the Gauss–Newton algorithm, and define the estimation error as

$$\Delta\boldsymbol{\eta} = \hat{\boldsymbol{\eta}} - \boldsymbol{\eta}^\star. \tag{73}$$

Our goal is to bound $\|\Delta\boldsymbol{\eta}\|_2$ in terms of the noise level ε and problem-dependent constants.

2) Local Linearization and Residual Expansion: Using a first-order Taylor expansion of the noiseless model $\boldsymbol{\mu}(\boldsymbol{\eta})$ around the true parameter vector $\boldsymbol{\eta}^\star$, we obtain

$$\boldsymbol{\mu}(\hat{\boldsymbol{\eta}}) = \boldsymbol{\mu}(\boldsymbol{\eta}^\star) + \mathbf{J}(\boldsymbol{\eta}^\star)\Delta\boldsymbol{\eta} + \mathbf{r}_{\text{nl}}, \tag{74}$$

where $\mathbf{J}(\boldsymbol{\eta}^\star)$ is the Jacobian matrix and $\mathbf{r}_{\text{nl}}$ collects higher-order terms.

The residual vector can be expressed as

$$\mathbf{r} = \boldsymbol{\mu}(\hat{\boldsymbol{\eta}}) - \mathbf{I} = \mathbf{J}(\boldsymbol{\eta}^\star)\Delta\boldsymbol{\eta} - \mathbf{n} + \mathbf{r}_{\text{nl}}. \tag{75}$$

3) Error Upper Bound via Inequality Relaxation: Assume that the Jacobian matrix satisfies a local identifiability condition, i.e.,

$$\sigma_{\min}\left(\mathbf{J}(\boldsymbol{\eta}^\star)\right) \ge \lambda_{\min} > 0, \tag{76}$$

where $\sigma_{\min}(\cdot)$ denotes the minimum singular value.

C. Precise Estimation Error Bound with Second-Order Residuals

1) From First-Order Approximation to Exact Expression with Second-Order Remainder: The starting point of the derivation is the first-order Taylor expansion of the residual function around the true parameter value $\boldsymbol{\eta}^\star$:

$$\mathbf{r}(\hat{\boldsymbol{\eta}}) = \mathbf{r}(\boldsymbol{\eta}^\star) + \mathbf{J}(\boldsymbol{\eta}^\star)\Delta\boldsymbol{\eta} + \boldsymbol{\Delta}_h, \tag{77}$$

where $\boldsymbol{\Delta}_h$ represents the higher-order remainder term, typically associated with the second-order Hessian. Rewriting this as:

$$\mathbf{J}(\boldsymbol{\eta}^\star)\Delta\boldsymbol{\eta} = \underbrace{\mathbf{r}(\hat{\boldsymbol{\eta}}) - \mathbf{r}(\boldsymbol{\eta}^\star)}_{\Delta\mathbf{r}} - \boldsymbol{\Delta}_h. \tag{78}$$

Substituting $\mathbf{r}(\boldsymbol{\eta}^\star) = \mathbf{n}$, we obtain the exact relationship:

$$\mathbf{J}(\boldsymbol{\eta}^\star)\Delta\boldsymbol{\eta} = \mathbf{r}(\hat{\boldsymbol{\eta}}) - \mathbf{n} - \boldsymbol{\Delta}_h. \tag{79}$$

2) Derivation of the Refined Error Bound: Taking the 2-norm of both sides of (79) and applying the triangle inequality yields:

$$\|\mathbf{J}(\boldsymbol{\eta}^\star)\Delta\boldsymbol{\eta}\|_2 = \|\mathbf{r}(\hat{\boldsymbol{\eta}}) - \mathbf{n} - \boldsymbol{\Delta}_h\|_2 \le \|\mathbf{r}(\hat{\boldsymbol{\eta}})\|_2 + \|\mathbf{n}\|_2 + \|\boldsymbol{\Delta}_h\|_2. \tag{80}$$

Utilizing the matrix norm property $\|\mathbf{J}\Delta\boldsymbol{\eta}\|_2 \ge \sigma_{\min}(\mathbf{J})\|\Delta\boldsymbol{\eta}\|_2$ and Assumption A1 ($\sigma_{\min}(\mathbf{J}(\boldsymbol{\eta}^\star)) \ge \lambda_{\min} > 0$), we have:

$$\lambda_{\min}\|\Delta\boldsymbol{\eta}\|_2 \le \|\mathbf{r}(\hat{\boldsymbol{\eta}})\|_2 + \|\mathbf{n}\|_2 + \|\boldsymbol{\Delta}_h\|_2. \tag{81}$$

Consequently, the estimation error bound is given by:

$$\boxed{\|\Delta\boldsymbol{\eta}\|_2 \le \frac{1}{\lambda_{\min}}\left(\|\mathbf{r}(\hat{\boldsymbol{\eta}})\|_2 + \|\mathbf{n}\|_2 + \|\boldsymbol{\Delta}_h\|_2\right)}. \tag{82}$$

3) Characterization of the Higher-Order Remainder: For practical applications, it is common to assume that the Hessian matrix is Lipschitz continuous, or to directly provide a conservative upper bound:

$$\|\boldsymbol{\Delta}_h\|_2 \le \frac{L}{2}\|\Delta\boldsymbol{\eta}\|_2^2, \tag{83}$$

where L is the Lipschitz constant of the function $\mathbf{h}$ within its domain (related to the magnitude of the second derivative).

4) Closed-Form Computable Error Bound: Substituting the remainder bound from (83) into (82) yields a closed-form inequality:

$$\|\Delta\boldsymbol{\eta}\|_2 \le \frac{1}{\lambda_{\min}}\left(\|\mathbf{r}(\hat{\boldsymbol{\eta}})\|_2 + \|\mathbf{n}\|_2\right) + \frac{L}{2\lambda_{\min}}\|\Delta\boldsymbol{\eta}\|_2^2. \tag{84}$$

This quadratic inequality in $\|\Delta\boldsymbol{\eta}\|_2$ provides a more refined characterization of the estimation error. For sufficiently small $\|\Delta\boldsymbol{\eta}\|_2$, the second-order term becomes negligible, leading to the simplified bound:

$$\|\Delta\boldsymbol{\eta}\|_2 \lessapprox \frac{1}{\lambda_{\min}}\left(\|\mathbf{r}(\hat{\boldsymbol{\eta}})\|_2 + \|\mathbf{n}\|_2\right). \tag{85}$$

The refined error bound in (82) explicitly decomposes the estimation error into three components: the optimization residual, the amplified noise, and the linearization error due to model nonlinearity. This decomposition provides clear insights into the factors influencing estimation accuracy and serves as a foundation for system design and analysis.

5) Error Bound for Symbol Estimation: During the data transmission phase, the symbol estimation problem admits a similar perturbation structure. Let $\mathbf{s}_t^\star$ and $\hat{\mathbf{s}}_t$ denote the true and estimated symbol vectors, respectively. Define $\Delta\mathbf{s}_t = \hat{\mathbf{s}}_t - \mathbf{s}_t^\star$.

Using a first-order expansion of $\boldsymbol{\mu}_t(\mathbf{s}_t)$ around $\mathbf{s}_t^\star$ yields

$$\boldsymbol{\mu}_t(\hat{\mathbf{s}}_t) = \boldsymbol{\mu}_t(\mathbf{s}_t^\star) + \mathbf{J}_s(\mathbf{s}_t^\star)\Delta\mathbf{s}_t + \mathbf{r}_{\mathrm{nl}}^{(s)}. \tag{86}$$

Assuming that $\sigma_{\min}(\mathbf{J}_s) \geq \lambda_s > 0$, we obtain

$$\|\Delta\mathbf{s}_t\|_2 \leq \frac{1}{\lambda_s}\left(\|\mathbf{r}_t\|_2 + \varepsilon\right), \tag{87}$$

which provides an explicit upper bound on the symbol estimation error.

6) Discussion and Limitations: The derived error bounds rely on local linearization and the conditioning of the Jacobian matrices. Several important observations follow:

- The bounds are *local* and hold when the Gauss–Newton iterations are initialized sufficiently close to the true parameters.
- The minimum singular value of the Jacobian is directly influenced by the presence of the reference wave and the spatial diversity of the RIS.
- In low-SNR regimes, higher-order terms may dominate, causing the bound to be loose.
- Unlike the CRB, the proposed bounds apply to a specific class of algorithms and therefore reflect algorithm-dependent performance guarantees.

D. Error Propagation and Performance Characterization

In this subsection, we analyze the estimation error behavior of the proposed two-stage framework by explicitly incorporating the RIS steering matrix $\mathbf{V}$ and the complex gain matrix $\mathbf{A}$ into the Gauss–Newton (GN) error model. Unlike generic algorithmic analyses, the following derivations reveal how system parameters such as noise power, RIS size, user number, and spatial separability jointly affect the estimation accuracy of both channel parameters and transmitted symbols.

1) Stage I: Error Analysis of $\mathbf{V}$ *and* $\mathbf{A}$ *Estimation:* Under the Gauss–Newton approximation, the estimation error after convergence is given by

$$\hat{\boldsymbol{\xi}} - \boldsymbol{\xi}_0 = \left(\mathbf{J}_\xi^\top \mathbf{J}_\xi\right)^{-1} \mathbf{J}_\xi^\top \mathbf{n}. \tag{88}$$

Assuming $\mathbb{E}[\mathbf{n}\mathbf{n}^\top] = \sigma_n^2 \mathbf{I}$, the mean squared estimation error satisfies

$$\mathbb{E}\left[\|\hat{\boldsymbol{\xi}} - \boldsymbol{\xi}_0\|_2^2\right] = \sigma_n^2 \operatorname{tr}\left((\mathbf{J}_\xi^\top \mathbf{J}_\xi)^{-1}\right) = \sigma_n^2 \sum_i \frac{1}{\sigma_i^2(\mathbf{J}_\xi)}, \tag{89}$$

where $\{\sigma_i(\mathbf{J}_\xi)\}$ denote the singular values of the Jacobian matrix.

Equation (89) reveals that the estimation accuracy of $\mathbf{V}$ and $\mathbf{A}$ is jointly determined by the noise power, the number of RIS elements, and the spatial distinguishability among users. In particular, increasing the RIS aperture enlarges the Jacobian dimension and improves its singular value profile, while closely spaced users lead to small singular values and degraded estimation accuracy.

2) Stage II: Symbol Estimation Error with Imperfect $\mathbf{V}$ *and* $\mathbf{A}$*:* In the data transmission stage, the received power measurements follow

$$\mathbf{I}^{(d)} = |\mathbf{V}\mathbf{A}\mathbf{s} + \mathbf{b}|^2 + \mathbf{n}, \tag{90}$$

where $\mathbf{s}$ denotes the unknown data symbol vector.

The receiver relies on the estimated matrices $\hat{\mathbf{V}}$ and $\hat{\mathbf{A}}$, leading to a model mismatch defined as

$$\Delta\mathbf{H} \triangleq \hat{\mathbf{V}}\hat{\mathbf{A}} - \mathbf{V}\mathbf{A}. \tag{91}$$

Linearizing the measurement model around the true symbol vector $\mathbf{s}_0$, the symbol estimation error can be approximated as

$$\hat{\mathbf{s}} - \mathbf{s}_0 \approx \left(\mathbf{J}_s^\top \mathbf{J}_s\right)^{-1} \mathbf{J}_s^\top \left(\mathbf{n} + 2\Re\{(\mathbf{V}\mathbf{A}\mathbf{s}_0 + \mathbf{b})^* \odot \Delta\mathbf{H}\mathbf{s}_0\}\right), \tag{92}$$

where $\mathbf{J}_s$ is the Jacobian matrix with respect to $\mathbf{s}$.

The resulting mean squared symbol estimation error is therefore governed by two additive components:

$$\mathbb{E}\left[\|\hat{\mathbf{s}} - \mathbf{s}_0\|_2^2\right] \lesssim \sigma_n^2 \sum_i \frac{1}{\sigma_i^2(\mathbf{J}_s)} + \|\Delta\mathbf{H}\|_F^2 \sum_i \frac{1}{\sigma_i^2(\mathbf{J}_s)}. \tag{93}$$

3) Discussion: Equation (93) explicitly demonstrates that the reliability of symbol detection is not only limited by the measurement noise but also by the accuracy of the estimated channel matrices. In particular, the conditioning and rank properties of the effective sensing matrix $\mathbf{VA}$ play a critical role. Increasing the number of RIS elements or improving angular separability enhances the smallest singular values of the Jacobian matrices in both stages, thereby simultaneously improving channel estimation accuracy and data detection performance. This analysis establishes a clear connection between physical system dimensions and achievable estimation reliability.

E. Discussion and Remarks

1) Physical Interpretation of $\lambda_{\min}$*:* In the context of this work, the parameter vector $\boldsymbol{\eta}$ consists of four physical quantities: the elevation angle θ, azimuth angle ϕ, real part of the channel complex gain $\alpha_\Re$, and imaginary part of the channel complex gain $\alpha_\Im$. The received vector $\mathbf{z}$ represents the power received at each element of the intelligent reflecting surface (IRS). The Jacobian matrix $\mathbf{J}(\boldsymbol{\eta}) = \partial\mathbf{h}(\boldsymbol{\eta})/\partial\boldsymbol{\eta}$ encodes the sensitivity of these received power measurements to variations in the parameters.

The minimum singular value $\lambda_{\min}$ of $\mathbf{J}(\boldsymbol{\eta}^\star)$ has a clear physical interpretation: it quantifies the *worst-case observability* of the parameter space directions. Specifically:

- When $\lambda_{\min}$ is large, all parameter directions are equally well-observable from the received power measurements, leading to accurate joint estimation of all four parameters.
- When $\lambda_{\min}$ approaches zero, it indicates that certain linear combinations of the parameters (particularly mixtures of angle and gain parameters) become nearly unobservable. This typically occurs in degenerate configurations, such as when the IRS elements are arranged in a co-linear fashion or when the incident signal arrives from certain

critical angles that create ambiguity between phase and amplitude variations.

- In practical IRS systems, $\lambda_{\min}$ is directly related to the *geometric diversity* of the array and the *angular separation* between potential source directions. A well-conditioned Jacobian ($\lambda_{\min} \gg 0$) ensures that small changes in any parameter produce distinguishable changes in the received power pattern across the array.

2) On the Initial Value Requirement of Taylor Expansion: The derivation in Section IV-C relies on the assumption that the estimation error $\|\Delta\boldsymbol{\eta}\|_2$ is sufficiently small to justify the first-order Taylor expansion. This requirement is indeed inherent to Gauss-Newton type optimization methods, which approximate nonlinear least-squares problems via local linearization.

It is important to clarify that this *local convergence* property is a characteristic of the optimization algorithm rather than a limitation specific to our analysis. Our error bound derivation provides a theoretical foundation for understanding the estimation accuracy *given that* the optimization converges to a neighborhood of the true solution. The study of global convergence guarantees and initialization strategies for such nonlinear estimation problems, while important, falls outside the scope of this paper, which focuses on characterizing the ultimate estimation precision achievable under proper convergence.

3) Note on the Lower Bound: A complementary Cramér-Rao lower bound (CRB) analysis for the same estimation problem has been presented elsewhere in this work (see Section X). The CRB provides the fundamental limit on the variance of any unbiased estimator, while the upper bound derived in Section IV-C characterizes the actual error of our specific estimator. Together, they form a complete picture: the CRB tells us how good the best possible estimator could be, while our upper bound tells us how good our particular estimator actually is.

4) Explicit Form of the Closed-Form Error Bound: The inequality in (84) provides an *implicit* bound for $\|\Delta\boldsymbol{\eta}\|_2$ since it appears on both sides. To obtain an *explicit* bound, we solve the quadratic inequality. Let $e = \|\Delta\boldsymbol{\eta}\|_2$, $A = \frac{1}{\lambda_{\min}}(\|\mathbf{r}(\hat{\boldsymbol{\eta}})\|_2 + \|\mathbf{n}\|_2)$, and $B = \frac{L}{2\lambda_{\min}}$. Then (84) becomes:

$$e \leq A + Be^2. \tag{94}$$

Assuming $4AB < 1$ (which corresponds to the case where the nonlinearity is sufficiently mild relative to the noise and residual), we can solve for e explicitly. The quadratic $Be^2 - e + A \geq 0$ has roots:

$$e_{1,2} = \frac{1 \pm \sqrt{1-4AB}}{2B}. \tag{95}$$

Since $e \geq 0$ and the quadratic opens upward, the inequality $e \leq A + Be^2$ holds for e between the two roots. The relevant explicit upper bound is the smaller root (as e must be less than or equal to this value to satisfy the inequality):

$$e \leq \frac{1 - \sqrt{1-4AB}}{2B}. \tag{96}$$

For small AB (i.e., when the nonlinear effects are weak relative to the linear term), we can expand (96) using the Taylor series $\sqrt{1-x} = 1 - \frac{x}{2} - \frac{x^2}{8} + \cdots$:

$$\begin{aligned} e &\leq \frac{1 - (1 - 2AB - 2A^2B^2 + \cdots)}{2B} \\ &= A + A^2B + 2A^3B^2 + \cdots \\ &= \frac{1}{\lambda_{\min}}(\|\mathbf{r}(\hat{\boldsymbol{\eta}})\|_2 + \|\mathbf{n}\|_2) + \frac{L}{2\lambda_{\min}^3}(\|\mathbf{r}(\hat{\boldsymbol{\eta}})\|_2 + \|\mathbf{n}\|_2)^2 + \mathcal{O}\left((\|\mathbf{r}(\hat{\boldsymbol{\eta}})\|_2 + \right. \end{aligned} \tag{97}$$

This explicit expansion reveals three important insights:

1) The leading term $\frac{1}{\lambda_{\min}}(\|\mathbf{r}(\hat{\boldsymbol{\eta}})\|_2 + \|\mathbf{n}\|_2)$ is the *linearized* error bound, which dominates when noise and residuals are small.
2) The second term $\frac{L}{2\lambda_{\min}^3}(\|\mathbf{r}(\hat{\boldsymbol{\eta}})\|_2 + \|\mathbf{n}\|_2)^2$ captures the *first-order effect of nonlinearity*, showing that the error increases super-linearly with noise/residual when nonlinearities are present.
3) The condition $4AB < 1$ translates to $\frac{2L}{\lambda_{\min}^2}(\|\mathbf{r}(\hat{\boldsymbol{\eta}})\|_2 + \|\mathbf{n}\|_2) < 1$, which defines the regime where the Taylor expansion remains valid and the bound is meaningful.

In practice, when the noise level and optimization residuals are sufficiently small, the linear term in (97) provides an adequate approximation, and the simplified bound (85) is both accurate and convenient for analysis.

5) Summary: The refined error bound developed in this section, along with the physical interpretation of $\lambda_{\min}$ and the explicit solution of the closed-form bound, provides a comprehensive theoretical framework for understanding the precision limits of parameter estimation in IRS-assisted systems. The analysis clarifies the interplay between array geometry (through $\lambda_{\min}$), measurement noise, optimization residuals, and system nonlinearities in determining ultimate estimation accuracy.

V. Performance Analysis of the Two stage Algorithm

A. Convergence Analysis of the Two stage Algorithm

This section establishes rigorous convergence properties of the proposed Gerchberg–Saxton (GS) type alternating projection algorithm used in the pilot phase to estimate the RIS steering matrix $\mathbf{V}(\boldsymbol{\theta})$ and the complex gain matrix $\mathbf{A}$ from power-only measurements.

Throughout this section, noise-free measurements are first assumed to clarify the geometric structure of the algorithm, and perturbation stability under noise is addressed subsequently.

1) Measurement Model and Feasible Sets: For the pilot stage, the received field at the RIS during snapshot t is

$$\mathbf{z}_t = \mathbf{b} + \mathbf{V}(\boldsymbol{\theta})\mathbf{A}\mathbf{s}_t \in \mathbb{C}^K, \tag{98}$$

where $\mathbf{b} \in \mathbb{C}^K$ denotes the known reference wave, $\mathbf{s}_t \in \mathbb{C}^L$ is the known pilot vector, $\mathbf{V}(\boldsymbol{\theta}) \in \mathbb{C}^{K\times L}$ is the RIS steering matrix, and $\mathbf{A} = \mathrm{diag}(\boldsymbol{\alpha})$ is the diagonal channel gain matrix.

The measured intensities are

$$\mathbf{y}_t = |\mathbf{z}_t|^2. \tag{99}$$

Stacking T snapshots gives

$$\mathbf{z} = \begin{bmatrix} \mathbf{z}_1 \\ \vdots \\ \mathbf{z}_T \end{bmatrix} \in \mathbb{C}^{KT}, \qquad \mathbf{y} = |\mathbf{z}|^2. \tag{100}$$

We define two feasible sets in the Hilbert space $\mathbb{C}^{KT}$.

$$\mathcal{A} = \left\{ \mathbf{z} \in \mathbb{C}^{KT} : |\mathbf{z}| = \sqrt{\mathbf{y}} \right\}. \tag{101}$$

$$\mathcal{B} = \left\{ \mathbf{z} : \exists(\boldsymbol{\theta}, \mathbf{A}) \text{ s.t. } \mathbf{z}_t = \mathbf{b} + \mathbf{V}(\boldsymbol{\theta})\mathbf{A}\mathbf{s}_t,\ \forall t \right\}. \tag{102}$$

where $\mathcal{A}$ is a non-convex set consisting of all complex vectors whose magnitudes match the measured intensities, and $\mathcal{B}$ is a smooth manifold parameterized by the physical parameters of interest.

The GS iteration alternates Euclidean projections between $\mathcal{A}$ and $\mathcal{B}$,

$$\mathbf{z}^{(k+1)} = P_{\mathcal{B}}\big(P_{\mathcal{A}}(\mathbf{z}^{(k)})\big). \tag{103}$$

2) Projection Operators: For any $\mathbf{z} \in \mathbb{C}^{KT}$, the projection onto $\mathcal{A}$ is given element-wise by

$$\big[P_{\mathcal{A}}(\mathbf{z})\big]_i = \sqrt{y_i}\frac{z_i}{|z_i|}, \qquad z_i \neq 0, \tag{104}$$

which satisfies

$$P_{\mathcal{A}}(\mathbf{z}) = \arg\min_{\mathbf{w}\in\mathcal{A}} \|\mathbf{w} - \mathbf{z}\|_2^2. \tag{105}$$

The projection onto $\mathcal{B}$ is defined as the solution of the structured least-squares problem

$$P_{\mathcal{B}}(\mathbf{z}) = \arg\min_{\tilde{\boldsymbol{\theta}},\tilde{\mathbf{A}}} \sum_{t=1}^{T} \big\|\mathbf{z}_t - \mathbf{b} - \mathbf{V}(\tilde{\boldsymbol{\theta}})\tilde{\mathbf{A}}\mathbf{s}_t\big\|_2^2. \tag{106}$$

For fixed $\boldsymbol{\theta}$, the minimizer in $\mathbf{A}$ admits the closed form

$$\hat{\mathbf{A}} = (\mathbf{V}^H\mathbf{V})^{-1}\mathbf{V}^H(\mathbf{Z} - \mathbf{b}\mathbf{1}^T)\mathbf{S}^H(\mathbf{S}\mathbf{S}^H)^{-1}, \tag{107}$$

provided that $\mathbf{S}\mathbf{S}^H$ and $\mathbf{V}^H\mathbf{V}$ are invertible, where $\mathbf{Z} = [\mathbf{z}_1, \ldots, \mathbf{z}_T]$ and $\mathbf{S} = [\mathbf{s}_1, \ldots, \mathbf{s}_T]$.

3) Monotonic Descent Property: We define the distance to a closed set $\mathcal{C}$ as

$$\mathrm{dist}(\mathbf{x}, \mathcal{C}) = \min_{\mathbf{y}\in\mathcal{C}} \|\mathbf{x} - \mathbf{y}\|_2. \tag{108}$$

Lemma 1 (Projection Inequality)**.** *For any closed set $\mathcal{C} \subset \mathbb{C}^n$,*

$$\|P_{\mathcal{C}}(\mathbf{x}) - \mathbf{y}\|_2^2 \leq \|\mathbf{x} - \mathbf{y}\|_2^2, \qquad \forall \mathbf{y} \in \mathcal{C}. \tag{109}$$

Proposition 1 (Monotonicity of GS)**.** *Let $\mathbf{z}^{(k+1)} = P_{\mathcal{B}}P_{\mathcal{A}}(\mathbf{z}^{(k)})$. Then*

$$\mathrm{dist}(\mathbf{z}^{(k+1)}, \mathcal{A}) \leq \mathrm{dist}(\mathbf{z}^{(k)}, \mathcal{A}), \tag{110}$$

$$\mathrm{dist}(\mathbf{z}^{(k+1)}, \mathcal{B}) \leq \mathrm{dist}(\mathbf{z}^{(k)}, \mathcal{B}). \tag{111}$$

Proof. Since $P_{\mathcal{A}}$ and $P_{\mathcal{B}}$ are Euclidean projections, the claim follows directly from the projection inequality applied sequentially to $\mathcal{A}$ and $\mathcal{B}$. □

4) Identifiability and Regularity Conditions: Let $(\boldsymbol{\theta}^\star, \mathbf{A}^\star)$ denote the true parameters and $\mathbf{z}^\star$ the corresponding stacked field.

We impose the following assumptions:

(C1) $\mathrm{rank}(\mathbf{S}) = L$.
(C2) $\sigma_{\min}(\mathbf{V}(\boldsymbol{\theta}^\star)) > 0$.
(C3) $K \geq 2L + 1$.

Under these conditions the mapping $(\boldsymbol{\theta}, \mathbf{A}) \mapsto \mathbf{z}$ is locally injective up to a global phase ambiguity.

Moreover, $\mathcal{B}$ is a smooth embedded manifold whose tangent space at $\mathbf{z}^\star$ is

$$\mathcal{T}_{\mathcal{B}}(\mathbf{z}^\star) = \mathrm{span}\left\{ \frac{\partial \mathbf{z}}{\partial \theta_\ell}, \frac{\partial \mathbf{z}}{\partial \alpha_\ell} \right\}_{\ell=1}^{L}. \tag{112}$$

5) Local Convergence:

Theorem 1 (Local Linear Convergence)**.** *Assume **(C1)–(C3)** hold and that the two manifolds $\mathcal{A}$ and $\mathcal{B}$ intersect transversally at $\mathbf{z}^\star$, i.e.,*

$$\mathcal{T}_{\mathcal{A}}(\mathbf{z}^\star) \cap \mathcal{T}_{\mathcal{B}}(\mathbf{z}^\star) = \{\mathbf{0}\}. \tag{113}$$

Then, for any initialization $\mathbf{z}^{(0)}$ sufficiently close to $\mathbf{z}^\star$, the GS iteration converges linearly to $\mathbf{z}^\star$,

$$\|\mathbf{z}^{(k)} - \mathbf{z}^\star\| \leq \rho^k \|\mathbf{z}^{(0)} - \mathbf{z}^\star\|, \qquad 0 < \rho < 1. \tag{114}$$

Proof. The result follows from the theory of alternating projections between smooth manifolds [**?**]. Transversality implies that the cosine of the Friedrichs angle between the tangent spaces is strictly less than one, which yields local linear convergence. □

6) Stability under Measurement Noise: Suppose that the observed powers satisfy

$$\mathbf{y} = |\mathbf{z}^\star|^2 + \mathbf{n}, \tag{115}$$

with $\|\mathbf{n}\| \leq \varepsilon$.

Then the amplitude set becomes perturbed,

$$\mathcal{A}_\varepsilon = \left\{ \mathbf{z} : \big||\mathbf{z}|^2 - \mathbf{y}\big| \leq \varepsilon \right\}. \tag{116}$$

By continuity of the projection operators, the GS iterates converge to a neighborhood of $\mathbf{z}^\star$ satisfying

$$\|\hat{\mathbf{z}} - \mathbf{z}^\star\| \leq C\varepsilon, \tag{117}$$

for some constant $C > 0$ depending on the local curvature of $\mathcal{B}$ and the intersection angle between $\mathcal{A}$ and $\mathcal{B}$.

B. Computational Complexity of the GS-Based Channel and DOA Estimation

In the first stage, the channel matrix $\mathbf{H} = \mathbf{V}(\boldsymbol{\theta})\mathbf{A} \in \mathbb{C}^{K\times L}$ and the associated angle parameters $\boldsymbol{\theta}$ are estimated from intensity-only measurements

$$\mathbf{y}_t = |\mathbf{H}\mathbf{s}_t + \mathbf{b}|^2 + \mathbf{n}_t, \qquad t = 1, \dots, T, \tag{118}$$

where K denotes the number of RIS elements, L the number of users, and T the pilot length.

The proposed GS-based solver alternates between enforcing the magnitude constraint and updating the channel estimate through a least-squares projection. In what follows, we derive the per-iteration computational complexity of this procedure.

1) Forward Propagation and Amplitude Projection: Given the current estimate $\mathbf{H}^{(i)}$, the forward propagation step computes

$$\mathbf{z}_t^{(i)} = \mathbf{H}^{(i)}\mathbf{s}_t + \mathbf{b}, \qquad t = 1, \dots, T. \tag{119}$$

This operation requires $O(KL)$ multiplications per snapshot, resulting in

$$O(KLT) \tag{120}$$

operations for all T pilots.

The amplitude projection step enforces the intensity constraint:

$$\tilde{\mathbf{z}}_t^{(i)} = \sqrt{\mathbf{y}_t} \odot \frac{\mathbf{z}_t^{(i)}}{|\mathbf{z}_t^{(i)}|}, \tag{121}$$

where $\odot$ denotes the Hadamard product. This step involves only element-wise operations and thus has complexity

$$O(KT). \tag{122}$$

Since typically $L \ll K$, the overall cost of this block is dominated by the forward propagation term $O(KLT)$.

2) Backward Projection via Least Squares: The backward projection updates the channel by solving

$$\mathbf{H}^{(i+1)} = \arg\min_{\mathbf{H}} \sum_{t=1}^{T} \left\| \mathbf{H}\mathbf{s}_t - \left(\tilde{\mathbf{z}}_t^{(i)} - \mathbf{b}\right) \right\|_2^2. \tag{123}$$

Defining

$$\mathbf{S} = [\mathbf{s}_1, \dots, \mathbf{s}_T] \in \mathbb{C}^{L\times T}, \quad \mathbf{Z}^{(i)} = [\tilde{\mathbf{z}}_1^{(i)} - \mathbf{b}, \dots, \tilde{\mathbf{z}}_T^{(i)} - \mathbf{b}] \in \mathbb{C}^{K\times T}, \tag{124}$$

the closed-form solution is

$$\mathbf{H}^{(i+1)} = \mathbf{Z}^{(i)}\mathbf{S}^\dagger, \tag{125}$$

where $\mathbf{S}^\dagger$ denotes the Moore–Penrose pseudoinverse.

If $\mathbf{S}$ is fixed throughout the pilot phase, its pseudoinverse can be precomputed at a cost of $O(L^2T + L^3)$ and reused for all iterations. Consequently, the dominant cost per iteration arises from the matrix multiplication

$$\mathbf{Z}^{(i)}\mathbf{S}^\dagger \in \mathbb{C}^{K\times L}, \tag{126}$$

which requires

$$O(KLT) \tag{127}$$

floating-point operations.

3) Incorporating the Parametric RIS Structure: In the considered RIS model, the channel is parameterized as

$$\mathbf{H} = \mathbf{V}(\boldsymbol{\theta})\mathbf{A}, \tag{128}$$

where $\mathbf{V}(\boldsymbol{\theta})$ consists of Kronecker steering vectors associated with the azimuth and elevation angles.

After obtaining $\mathbf{H}^{(i+1)}$ from the least-squares step, the angle parameters $\boldsymbol{\theta}$ and gains $\mathbf{A}$ are refined by projecting $\mathbf{H}^{(i+1)}$ onto the structured manifold $\{\mathbf{V}(\boldsymbol{\theta})\mathbf{A}\}$, for example through nonlinear least squares:

$$\min_{\boldsymbol{\theta},\mathbf{A}} \left\| \mathbf{H}^{(i+1)} - \mathbf{V}(\boldsymbol{\theta})\mathbf{A} \right\|_F^2. \tag{129}$$

A Gauss–Newton or gradient-based update of this problem requires evaluating steering vectors and Jacobians of dimension $K \times L$, leading to an additional complexity on the order of

$$O(KL) \tag{130}$$

per inner iteration, which is negligible compared with the dominant $O(KLT)$ cost when T is moderately large.

4) Overall Per-Iteration Complexity: Collecting the above results, the per-iteration complexity of the GS-based channel and DOA estimation procedure is given by

$$\boxed{\mathcal{O}(KLT)} \tag{131}$$

floating-point operations.

Let I_{GS} denote the number of GS iterations required to reach convergence. The total computational burden in the pilot phase therefore scales as

$$\mathcal{O}(I_{\text{GS}}KLT). \tag{132}$$

This cubic dependence on the system dimensions reflects the fact that the GS algorithm is applied to a high-dimensional, nonlinear phase-retrieval problem involving all RIS elements and pilot snapshots.

C. Convergence Analysis of Gauss–Newton Symbol Estimation

In the second stage, the channel matrix and steering vectors are assumed to be accurately estimated from the pilot phase. The received intensity measurements at the RIS are modeled as

$$\mathbf{y} = |\mathbf{V}\mathbf{A}\mathbf{s} + \mathbf{b}|^2 + \mathbf{n}, \tag{133}$$

where $\mathbf{V} \in \mathbb{C}^{K\times L}$ is the steering matrix, $\mathbf{A} = \text{diag}(\alpha_1, \dots, \alpha_L)$ is the diagonal channel gain matrix, $\mathbf{b} \in \mathbb{C}^K$ is the known reference wave, and $\mathbf{s} \in \mathbb{C}^L$ denotes the transmitted QPSK symbol vector to be recovered.

Define the noiseless nonlinear mapping

$$\boldsymbol{\mu}(\mathbf{s}) = |\mathbf{VAs} + \mathbf{b}|^2 . \tag{134}$$

Symbol recovery is formulated as the nonlinear least–squares problem

$$\min_{\mathbf{s}} \; f(\mathbf{s}) = \frac{1}{2} \left\| \boldsymbol{\mu}(\mathbf{s}) - \mathbf{y} \right\|_2^2 . \tag{135}$$

To apply real–valued optimization, we introduce the parameter vector

$$\mathbf{x} = \begin{bmatrix} \Re\{\mathbf{s}\} \\ \Im\{\mathbf{s}\} \end{bmatrix} \in \mathbb{R}^{2L}. \tag{136}$$

Let

$$\mathbf{z}(\mathbf{s}) = \mathbf{VAs} + \mathbf{b}, \tag{137}$$

and define the residual vector

$$\mathbf{r}(\mathbf{x}) = |\mathbf{z}(\mathbf{s})|^2 - \mathbf{y}. \tag{138}$$

For the kth RIS element, the residual is

$$r_k = |z_k|^2 - y_k. \tag{139}$$

Using Wirtinger calculus and real parameterization, the Jacobian matrix of $\mathbf{r}(\mathbf{x})$ with respect to $\mathbf{x}$ can be written as

$$\mathbf{J}(\mathbf{x}) = 2\,\Re\Big(\mathrm{diag}\big(\mathbf{z}^*(\mathbf{s})\big)\mathbf{VA}\Big) \in \mathbb{R}^{K\times 2L}. \tag{140}$$

The Gauss–Newton (GN) iteration for solving (135) is therefore

$$\mathbf{x}^{(i+1)} = \mathbf{x}^{(i)} - \big(\mathbf{J}^T\mathbf{J}\big)^{-1}\mathbf{J}^T\mathbf{r}. \tag{141}$$

The convergence of the GN method follows from classical results in nonlinear least–squares theory and the Newton–Kantorovich theorem.

Theorem 2 (Local Quadratic Convergence)**.** *Suppose that in the noiseless case $\mathbf{y} = \boldsymbol{\mu}(\mathbf{s}^\star)$, the following conditions hold:*

1) *The Jacobian matrix $\mathbf{J}(\mathbf{s}^\star)$ has full column rank;*
2) $\mathbf{J}(\mathbf{s})$ *is locally Lipschitz continuous in a neighborhood of $\mathbf{s}^\star$, i.e.,*

$$\|\mathbf{J}(\mathbf{s}_1) - \mathbf{J}(\mathbf{s}_2)\| \leq L_J \|\mathbf{s}_1 - \mathbf{s}_2\|; \tag{142}$$

3) *The initial point satisfies*

$$\|\mathbf{s}^{(0)} - \mathbf{s}^\star\| \leq \frac{\sigma_{\min}(\mathbf{J}(\mathbf{s}^\star))}{L_J}. \tag{143}$$

Then the GN iterates (141) *satisfy*

$$\|\mathbf{s}^{(i+1)} - \mathbf{s}^\star\| \leq C\|\mathbf{s}^{(i)} - \mathbf{s}^\star\|^2 \tag{144}$$

for some constant $C > 0$.

Proof: The exact Hessian of $f(\mathbf{s})$ is

$$\nabla^2 f = \mathbf{J}^T\mathbf{J} + \sum_{k=1}^{K} r_k \nabla^2 r_k. \tag{145}$$

Near the optimum, $r_k \approx 0$, so the Hessian is well approximated by $\mathbf{J}^T\mathbf{J}$. Under the stated Lipschitz and rank conditions, the Newton–Kantorovich theorem guarantees local quadratic convergence of the GN iterates.

The presence of the reference wave $\mathbf{b}$ plays a crucial role in conditioning the Jacobian matrix. From (140), each row of $\mathbf{J}$ scales with z_k^*. When $\mathbf{b} = \mathbf{0}$ and $\mathbf{VAs}$ contains small entries, $\mathbf{J}$ may become ill–conditioned or rank–deficient.

With a nonzero reference wave, however,

$$z_k \approx b_k, \tag{146}$$

which yields the bound

$$\sigma_{\min}\big(\mathbf{J}\big) \geq 2 \min_k |b_k| \; \sigma_{\min}(\mathbf{VA}). \tag{147}$$

Hence, the reference wave ensures stable inversion of the normal equations in (141).

In the presence of additive noise, linearizing the residual equation around $\mathbf{s}^\star$ gives

$$\delta\mathbf{s} = \big(\mathbf{J}^T\mathbf{J}\big)^{-1}\mathbf{J}^T\mathbf{n}. \tag{148}$$

Consequently,

$$\mathbb{E}\{\delta\mathbf{s}\,\delta\mathbf{s}^T\} \succeq \sigma^2(\mathbf{J}^T\mathbf{J})^{-1}, \tag{149}$$

which coincides with the Cramér–Rao lower bound (CRB).

Moreover,

$$\|\delta\mathbf{s}\| \leq \frac{\|\mathbf{n}\|}{\sigma_{\min}(\mathbf{J})}, \tag{150}$$

highlighting again the importance of the conditioning of $\mathbf{VA}$ and the reference wave.

D. Computational Complexity

Let K denote the number of RIS elements and L the number of users.

At each GN iteration:

- residual and Jacobian evaluation: $\mathcal{O}(KL)$;
- forming $\mathbf{J}^T\mathbf{J}$: $\mathcal{O}(KL^2)$;
- solving the normal equations: $\mathcal{O}(L^3)$.

Thus, the total complexity is

$$\mathcal{O}\big(I_{\mathrm{GN}}\left(KL^2 + L^3\right)\big), \tag{151}$$

where I_{GN} is the number of GN iterations.

Compared with the Gerchberg–Saxton algorithm employed in the channel estimation stage, which typically requires a much larger number of iterations and repeated forward–backward projections over K RIS elements and T pilot symbols, the GN method is considerably more efficient for real–time symbol detection.

E. Summary: Rationale for the Two-Stage GS–GN Estimation Strategy

We finally summarize the theoretical motivations for adopting a two-stage estimation strategy, in which the channel and wavevector parameters are recovered via a gradient-based scheme, while the transmitted symbols are refined through a Gauss–Newton (GN) iteration.

Computational Complexity Comparison: Let N denote the number of RIS elements, M the number of pilot observations, and K the number of unknown channel or angular parameters. In the first stage, the GS algorithm updates the channel-related vector $\boldsymbol{\theta} \in \mathbb{C}^K$ by a projected gradient step of the form

$$\boldsymbol{\theta}^{(t+1)} = \mathcal{P}_{\mathcal{C}}\left(\boldsymbol{\theta}^{(t)} - \mu_t \nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}^{(t)})\right), \tag{152}$$

where $\mathcal{L}$ is a nonlinear least-squares cost function associated with the magnitude-only measurements.

As derived previously, one GS iteration mainly consists of

- evaluating forward operators of size $M \times K$,
- computing Wirtinger gradients,
- and performing simple Euclidean or manifold projections.

Hence, the per-iteration complexity scales as

$$\mathcal{O}(MK + K) \approx \mathcal{O}(MK), \tag{153}$$

which is linear in the parameter dimension.

In contrast, in the second stage the GN algorithm is applied to refine the transmitted symbol vector $\mathbf{s} \in \mathbb{C}^Q$ under a known channel estimate $\widehat{\mathbf{H}}$, by solving at iteration t

$$\mathbf{s}^{(t+1)} = \mathbf{s}^{(t)} - \left(\mathbf{J}^H \mathbf{J}\right)^{-1} \mathbf{J}^H \mathbf{r}(\mathbf{s}^{(t)}), \tag{154}$$

where $\mathbf{r}(\mathbf{s})$ is the residual vector and $\mathbf{J}$ is its Jacobian matrix.

The dominant computational burden is the formation and inversion of the normal matrix $\mathbf{J}^H\mathbf{J} \in \mathbb{C}^{Q \times Q}$, yielding a complexity of

$$\mathcal{O}\big(MQ^2 + Q^3\big), \tag{155}$$

which is significantly higher than that of GS when Q is moderate or large.

Therefore, from a computational standpoint, GS is more suitable for handling the high-dimensional channel and angular parameters in the first stage, whereas GN is reserved for the second stage where the unknown dimension is smaller but high accuracy is required.

Convergence-Order Comparison: Beyond computational cost, the fundamental distinction between the two algorithms lies in their convergence rates.

Under standard smoothness assumptions on $\mathcal{L}(\boldsymbol{\theta})$ and a sufficiently small stepsize μ_t, projected GS satisfies the descent inequality

$$\mathcal{L}(\boldsymbol{\theta}^{(t+1)}) \leq \mathcal{L}(\boldsymbol{\theta}^{(t)}) - c\left\|\nabla \mathcal{L}(\boldsymbol{\theta}^{(t)})\right\|^2, \tag{156}$$

for some constant $c > 0$, and converges to a stationary point with at most a *linear* rate in a local neighborhood,

$$\|\boldsymbol{\theta}^{(t+1)} - \boldsymbol{\theta}^\star\| \leq \rho\|\boldsymbol{\theta}^{(t)} - \boldsymbol{\theta}^\star\|, \qquad 0 < \rho < 1. \tag{157}$$

In contrast, if the residual function $\mathbf{r}(\mathbf{s})$ is twice continuously differentiable and the Jacobian $\mathbf{J}$ has full column rank at the solution $\mathbf{s}^\star$, classical nonlinear least-squares theory guarantees that GN achieves *quadratic local convergence*, namely,

$$\|\mathbf{s}^{(t+1)} - \mathbf{s}^\star\| \leq C\|\mathbf{s}^{(t)} - \mathbf{s}^\star\|^2, \tag{158}$$

for some constant $C > 0$ when $\mathbf{s}^{(t)}$ is sufficiently close to $\mathbf{s}^\star$.

This fundamental difference in convergence order implies that GN requires far fewer iterations than GS to reach a high-precision estimate once a reliable initialization is available.

Implications for the Proposed Two-Stage Estimation Framework: Combining the above complexity and convergence analyses, the proposed two-stage strategy is theoretically justified as follows:

- The first-stage GS algorithm efficiently handles the large-scale, nonconvex channel and wavevector estimation problem with a low per-iteration cost and guaranteed monotonic descent.
- The second-stage GN algorithm exploits the refined channel estimate to perform rapid, high-accuracy symbol recovery thanks to its quadratic local convergence.

Hence, although GN is computationally more expensive per iteration, its superior convergence order makes it the preferable choice for symbol refinement, whereas GS remains optimal for the high-dimensional channel inference stage. This complementary behavior establishes the optimality of employing GS and GN in two separate phases rather than forcing a single algorithmic paradigm to address both estimation tasks.

VI. Single-Snapshot Identifiability With RIS Power Measurements

A. Stage I: Channel and DOA Estimation Model

We consider a reconfigurable intelligent surface (RIS) composed of $M \times N$ reflecting elements located on the yoz-plane. Let $K = MN$ denote the total number of elements. For the ℓ-th user, the RIS steering vector is given by the Kronecker product

$$\mathbf{v}_\ell = \mathbf{a}_z(\theta_\ell, \phi_\ell) \otimes \mathbf{a}_y(\theta_\ell, \phi_\ell) \in \mathbb{C}^K, \tag{159}$$

where

$$\mathbf{a}_y(\theta, \phi) = \left[e^{jk_y(\theta,\phi)y_1}, \ldots, e^{jk_y(\theta,\phi)y_N}\right]^\top, \tag{160}$$

$$\mathbf{a}_z(\theta, \phi) = \left[e^{jk_z(\theta,\phi)z_1}, \ldots, e^{jk_z(\theta,\phi)z_M}\right]^\top. \tag{161}$$

Collecting the L users, the RIS channel matrix is

$$\mathbf{H} = [\mathbf{h}_1, \ldots, \mathbf{h}_L] = [\alpha_1\mathbf{v}_1, \ldots, \alpha_L\mathbf{v}_L] = \mathbf{V}(\boldsymbol{\Theta}, \boldsymbol{\Phi})\mathbf{A}, \tag{162}$$

where $\mathbf{V} = [\mathbf{v}_1, \ldots, \mathbf{v}_L]$, $\mathbf{A} = \mathrm{diag}(\alpha_1, \ldots, \alpha_L)$, and (θ_ℓ, ϕ_ℓ) denotes the direction-of-arrival (DoA) of the ℓ-th user.

During one pilot symbol interval, the RIS measures only the received signal magnitudes,

$$\mathbf{I} = \left|\mathbf{H}\mathbf{x} + \mathbf{b}\right| = \left|\sum_{\ell=1}^{L} \alpha_\ell x_\ell \mathbf{v}_\ell + \mathbf{b}\right|, \tag{163}$$

where $\mathbf{x} \in \mathbb{C}^L$ is the known pilot vector, $\mathbf{b} \in \mathbb{C}^K$ is a known reference wave, and $|\cdot|$ is applied elementwise.

Defining $\beta_\ell \triangleq \alpha_\ell x_\ell$ and $\boldsymbol{\beta} = [\beta_1, \ldots, \beta_L]^\top$, (163) becomes

$$\mathbf{I} = \left|\mathbf{V}(\boldsymbol{\Theta}, \boldsymbol{\Phi})\boldsymbol{\beta} + \mathbf{b}\right|. \tag{164}$$

The unknown real-valued parameter vector is

$$\boldsymbol{\psi} = \{\theta_\ell, \phi_\ell, \Re\{\beta_\ell\}, \Im\{\beta_\ell\}\}_{\ell=1}^L, \tag{165}$$

which has dimension $4L$.

1) Jacobian of the Magnitude Mapping: Let

$$\mathbf{s} = \mathbf{V}\boldsymbol{\beta} + \mathbf{b}, \qquad I_k = |s_k|. \tag{166}$$

For $s_k \neq 0$, Wirtinger calculus yields

$$\frac{\partial I_k}{\partial \boldsymbol{\psi}} = \Re\left\{ \frac{s_k^*}{|s_k|} \frac{\partial s_k}{\partial \boldsymbol{\psi}} \right\}. \tag{167}$$

Stacking all $k = 1, \ldots, K$ gives the Jacobian matrix

$$\mathbf{J}(\boldsymbol{\psi}) = \Re\left\{ \mathbf{D}(\mathbf{s}) \frac{\partial(\mathbf{V}\boldsymbol{\beta})}{\partial \boldsymbol{\psi}} \right\}, \tag{168}$$

where

$$\mathbf{D}(\mathbf{s}) = \mathrm{diag}\left(\frac{s_1^*}{|s_1|}, \ldots, \frac{s_K^*}{|s_K|} \right). \tag{169}$$

For the ℓ-th user, the partial derivatives satisfy

$$\frac{\partial(\mathbf{V}\boldsymbol{\beta})}{\partial \beta_\ell} = \mathbf{v}_\ell, \tag{170}$$

$$\frac{\partial(\mathbf{V}\boldsymbol{\beta})}{\partial \theta_\ell} = \beta_\ell \frac{\partial \mathbf{v}_\ell}{\partial \theta_\ell}, \tag{171}$$

$$\frac{\partial(\mathbf{V}\boldsymbol{\beta})}{\partial \phi_\ell} = \beta_\ell \frac{\partial \mathbf{v}_\ell}{\partial \phi_\ell}. \tag{172}$$

For a two-dimensional Vandermonde array manifold, the vectors $\mathbf{v}_\ell$, $\partial\mathbf{v}_\ell/\partial\theta_\ell$, and $\partial\mathbf{v}_\ell/\partial\phi_\ell$ are linearly independent for generic angles.

2) Local Identifiability: The following result characterizes when the DoAs and complex gains can be uniquely determined from a single RIS snapshot.

Theorem 3 (Single-Snapshot Identifiability)**.** *Consider the measurement model* (164)*. Assume that*

1) *The RIS geometry is non-degenerate and planar.*
2) *The angles satisfy $(\theta_\ell, \phi_\ell) \neq (\theta_m, \phi_m)$ for $\ell \neq m$.*
3) *$\beta_\ell \neq 0$ for all ℓ.*
4) *There exists an index set $\mathcal{K}$ with $|\mathcal{K}| \geq 4L$ such that $s_k \neq 0$ for all $k \in \mathcal{K}$.*
5) *The number of RIS elements satisfies $K \geq 4L$.*

Then the mapping $(\boldsymbol{\Theta}, \boldsymbol{\Phi}, \boldsymbol{\beta}) \mapsto |\mathbf{V}\boldsymbol{\beta} + \mathbf{b}|$ is locally injective, i.e., all DoAs and complex gains are uniquely determined (up to isolated degenerate cases) from a single snapshot.

Proof. Local injectivity follows if the Jacobian $\mathbf{J}(\boldsymbol{\psi})$ in (168) has full column rank $4L$. Conditions 1)–2) ensure that $\mathbf{V}$ is full column rank and that $\mathbf{v}_\ell$, $\partial\mathbf{v}_\ell/\partial\theta_\ell$, and $\partial\mathbf{v}_\ell/\partial\phi_\ell$ span a three-dimensional subspace for each ℓ. Condition 3) prevents the angular derivatives from vanishing. Condition 4) guarantees that the diagonal matrix $\mathbf{D}(\mathbf{s})$ is well defined and nonsingular over $\mathcal{K}$, which removes the global phase ambiguity typical in phaseless measurements. Finally, Condition 5) ensures that the number of real observations is no smaller than the number of unknown parameters. Consequently, the $4L$ columns of $\mathbf{J}(\boldsymbol{\psi})$ are linearly independent, implying that the mapping is locally one-to-one. □

Remark 1. *The full-rank Jacobian also implies that the Gauss–Newton algorithm applied to* (164) *is locally convergent and that the normal matrix $\mathbf{J}^\top\mathbf{J}$ is positive definite at the true solution.*

B. Stage II: Symbol Detection

C. Geometric Identifiability of Symbols via a Reference Wave

We consider the single–measurement magnitude model

$$y = |b + hs|, \tag{173}$$

where $b \in \mathbb{C}$ denotes a known reference wave, $h \in \mathbb{C}$ is the effective channel coefficient, and s is drawn from a finite constellation $\mathcal{S}$. Throughout this subsection, the channel h is assumed to be perfectly known, which corresponds to the second-stage symbol estimation scenario.

1) Amplitude Mapping and Injectivity: Define the nonlinear mapping

$$f : \mathcal{S} \to \mathbb{R}_+, \qquad f(s) \triangleq |b + hs|. \tag{174}$$

Recovering s from a single magnitude-only observation y is possible if and only if $f(\cdot)$ is injective over $\mathcal{S}$, i.e.,

$$f(s_i) \neq f(s_j), \qquad \forall s_i \neq s_j. \tag{175}$$

Geometrically, b and hs can be interpreted as vectors in the complex plane. For each $s \in \mathcal{S}$, the composite vector $z_s = b + hs$ originates from the origin and terminates at a constellation-dependent point, while $f(s)$ equals its Euclidean distance to the origin. Condition (175) therefore requires that all z_s lie on circles of distinct radii.

2) Closed-Form Expression of the Magnitude: Let

$$b = |b|e^{j\phi_b}, \qquad h = |h|e^{j\phi_h}, \tag{176}$$

and write $s = e^{j\phi_s}$ for constant-modulus constellations such as PSK. Then

$$\begin{aligned} |b + hs|^2 &= (b + hs)(b^* + h^*s^*) \\ &= |b|^2 + |h|^2 + 2\Re\{b^*hs\} \\ &= |b|^2 + |h|^2 + 2|b||h|\cos(\phi_h + \phi_s - \phi_b). \end{aligned} \tag{177}$$

Define

$$\theta_s \triangleq \phi_h + \phi_s - \phi_b. \tag{178}$$

Then the amplitude mapping is fully characterized by

$$f^2(s) = |b|^2 + |h|^2 + 2|b||h|\cos\theta_s. \tag{179}$$

3) Identifiability for QPSK: For QPSK,

$$\mathcal{S} = \left\{ e^{j\frac{\pi}{4}}, e^{j\frac{3\pi}{4}}, e^{j\frac{5\pi}{4}}, e^{j\frac{7\pi}{4}} \right\}. \tag{180}$$

From (179), injectivity is equivalent to

$$\cos\theta_{s_i} \neq \cos\theta_{s_j}, \qquad \forall s_i \neq s_j. \tag{181}$$

Since $\cos x = \cos y$ if and only if $x = \pm y + 2k\pi$, $k \in \mathbb{Z}$, identifiability fails only when the relative phase $\phi_h - \phi_b$ aligns with a symmetry axis of the QPSK constellation. These degenerate cases form a measure-zero subset of $\mathbb{R}$. Therefore, if (b, h) are drawn from any continuous distribution, the injectivity condition (181) holds almost surely.

4) Minimum Amplitude Separation and Reference-Wave Design: Beyond mere injectivity, detection robustness in noise depends on the minimum amplitude separation

$$\Delta_{\min}(b) \triangleq \min_{s_i \neq s_j} \big|\, |b + hs_i| - |b + hs_j| \,\big| . \tag{182}$$

Maximizing $\Delta_{\min}(b)$ enlarges the decision margins of amplitude-based detectors and directly improves the symbol error probability.

Using (179), the squared-amplitude difference between two symbols is

$$f^2(s_i) - f^2(s_j) = 2|b||h|\Big(\cos\theta_{s_i} - \cos\theta_{s_j}\Big). \tag{183}$$

Hence, $\Delta_{\min}(b)$ is jointly controlled by:

- the magnitude $|b|$ of the reference wave;
- the channel gain $|h|$;
- the phase offset ϕ_b relative to ϕ_h.

For fixed h, an effective design principle is to select ϕ_b such that the angles $\{\theta_s\}_{s\in\mathcal{S}}$ are maximally separated on $[0, 2\pi)$, i.e.,

$$\phi_b^\star = \arg\max_{\phi_b} \min_{s_i \neq s_j} \big|\cos\theta_{s_i} - \cos\theta_{s_j}\big|. \tag{184}$$

Similarly, the magnitude $|b|$ controls the overall radial stretching of the composite constellation and should be chosen to avoid the degenerate regimes $|b| \ll |h|$ or $|b| \gg |h|$, in which the amplitude differences become vanishingly small.

5) Discussion: The above geometric analysis reveals that the introduction of a properly designed reference wave transforms the phase-ambiguous magnitude-only observation into an injective mapping over finite constellations. This property provides a fundamental explanation for the improved symbol identifiability observed in the second stage of the proposed framework and motivates explicit reference-wave design strategies in RIS-assisted noncoherent communications. ■

VII. Simulation Results and Analysis

A. Simulation Setup

We evaluate the proposed algorithm through MATLAB simulations with the following parameters:

- RIS configuration: $N = 8$, $M = 8$, total elements $K = 64$
- Transmitted signal: $s = 0.8 + j0.6$ (magnitude 1.0, phase 0.6435 rad)
- Channel: Rayleigh fading, $h_i \sim \mathcal{CN}(0, 1)$
- Reference wave: $b_i = e^{j\theta_i}$ with $\theta_i \sim \mathcal{U}(0, 2\pi)$
- Signal-to-noise ratio (SNR): varied from 0 to 30 dB
- Initialization: random magnitude in $[0, 2]$, random phase in $[0, 2\pi]$
- Convergence tolerance: $\epsilon = 10^{-6}$

TABLE I: Performance comparison of signal recovery algorithms

Algorithm	NMSE (dB)	Runtime (ms)	Convergence Rate
Gauss-Newton (proposed)	-25.3	4.2	98%
Levenberg-Marquardt	-25.1	5.8	100%
Alternating Projection	-18.7	12.5	85%
Wirtinger Flow	-22.4	8.3	92%
Semidefinite Relaxation	-24.8	45.6	100%

B. Performance Metrics

The performance is evaluated using:

- Normalized mean squared error (NMSE): NMSE $= \frac{\mathbb{E}[|s-\hat{s}|^2]}{|s|^2}$
- Magnitude error: $\frac{\mathbb{E}[||s|-|\hat{s}||]}{|s|}$
- Phase error: $\mathbb{E}[\min(|\angle s - \angle\hat{s}|, 2\pi - |\angle s - \angle\hat{s}|)]$
- Convergence rate: percentage of trials converging within maximum iterations

Figure **??** shows the typical convergence behavior of the Gauss-Newton algorithm. The objective function decreases monotonically, with rapid convergence in the initial iterations followed by refinement in later iterations. The algorithm typically converges within 10-20 iterations for moderate SNR conditions.

Figure **??** presents the distribution of magnitude and phase errors across 1000 independent trials. The magnitude error is concentrated below 5%, while the phase error is typically less than 0.1 rad, demonstrating the algorithm's accuracy.

Figure **??** illustrates the relationship between NMSE and SNR. The algorithm approaches the theoretical lower bound at high SNR, with a 3 dB loss at low SNR due to the nonlinear measurement model. The performance improvement with increasing SNR validates the algorithm's noise resilience.

C. Comparison with Alternative Methods

Table I compares the proposed Gauss-Newton approach with alternative methods:

- **Alternating Projection**: Phase retrieval via Gerchberg-Saxton algorithm
- **Wirtinger Flow**: Gradient descent on amplitude loss
- **Semidefinite Relaxation**: Convex relaxation via lifting

The proposed Gauss-Newton algorithm achieves the best NMSE performance with the shortest runtime, demonstrating its efficiency and effectiveness for this problem.

VIII. Conclusion

This paper presented a comprehensive framework for signal recovery in holographic RIS receivers using only amplitude measurements. By employing a known reference wave and formulating the recovery as a nonlinear least-squares problem, we enable the estimation of both amplitude and phase of the transmitted signal. The proposed Gauss-Newton algorithm efficiently solves this optimization, with simulations demonstrating accurate recovery under various SNR conditions.

The mathematical derivation provides complete details from signal transmission through the channel, interference with the reference wave at the RIS, intensity measurement, to the final recovery algorithm. This framework serves as a foundation for more advanced RIS receiver designs, including those with adaptive reflection coefficients, multiple users, and frequency-selective channels.

Future work will extend this approach to practical implementations with hardware impairments, investigate deep learning-based recovery for reduced complexity, and explore applications in joint communication and sensing with RIS.

Appendix A

Appendix B

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